

Remote Sensing and GNSS-based Validation for Environmental Monitoring

Case Studies from Estonia

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Estonia

Capital: Tallinn

Population: 1.3 million

Area: 45 000 km²

Language: Estonian

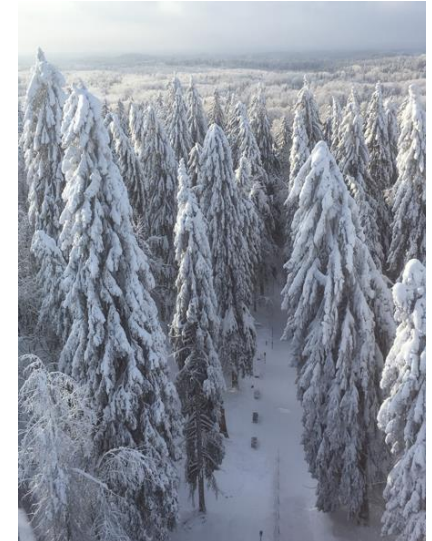
Government: Parliamentary republic

Member: European Union and NATO from 2004

Currency: Euro €



- More than 50% of the country is covered by forests
- Over 2,000 islands and islets
- Coastline $\approx$ 3,800 km
- Highest point: 318 m



Estonians

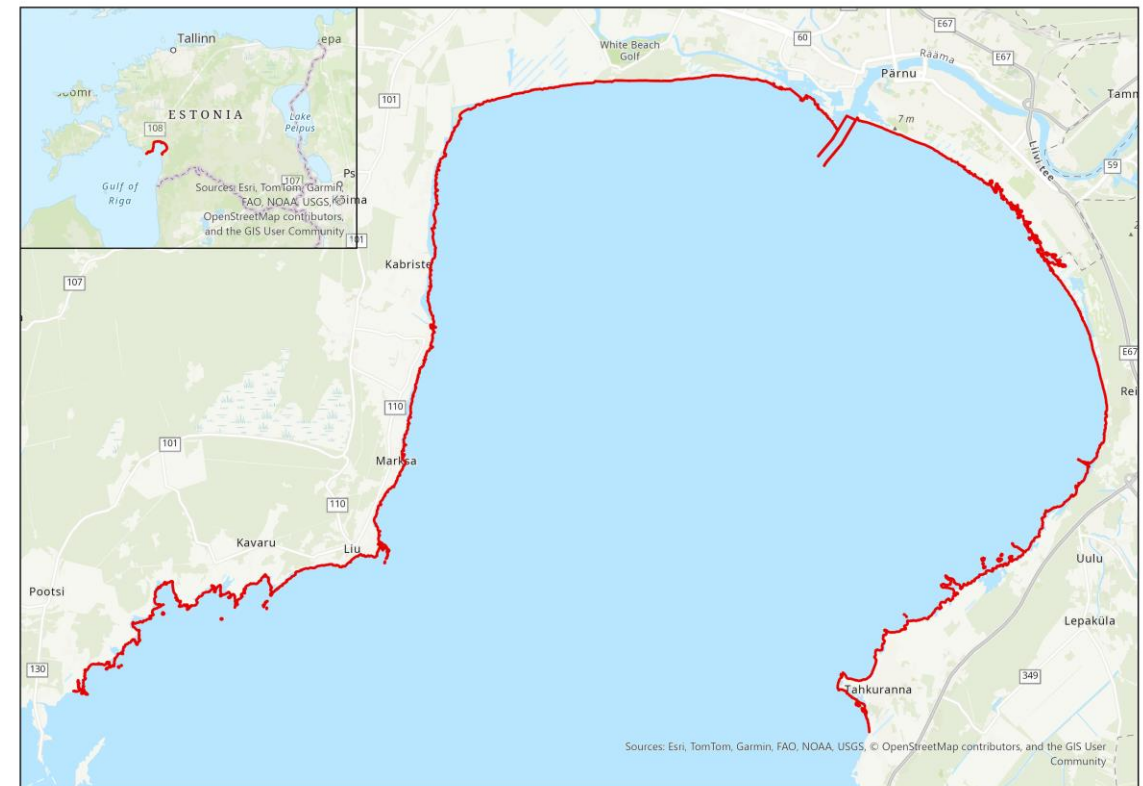
- Strong connection to nature
- Independent-minded
- Reserved
- Sauna people
- Singing nation
- Digital innovators



First Case Study

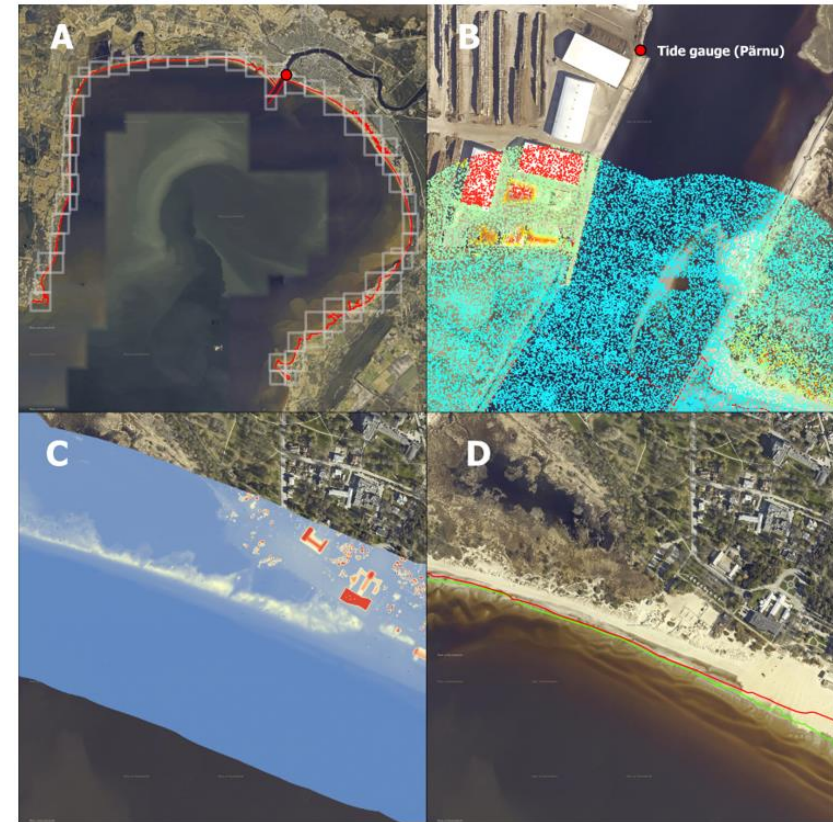
Coastline Mapping Using Sentinel-1 and Sentinel-2 Data

- What is the reference level for estimating land loss due to global sea-level rise?
- In Estonia:
 - Aerial photos
 - Airborne Laser Scanning (ALS)
 - Tide gauges (TG) time series (2010–2020)



Data

- Official coastline (reference coastline)
- ALS point clouds: May 2021
- Sentinel-2 Level-2A imagery (2015–2025, March–Nov.), 10 m resolution
- Sentinel-1 GRD (Ground Range Detected) imagery (2015–2025, March–Nov.), 10 m resolution



A – 1: 2000 map sheets

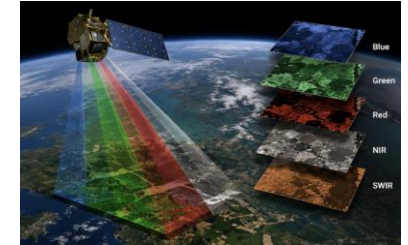
B – ALS point cloud

C – DEM

D – Green: coastline based on DEM at 0.24 m (MSL for 2015–2025);
Red: reference coastline



Sentinel-1 and Sentinel-2



Sentinel-1 (SAR, VH polarization)

Input: VH-polarized backscatter

Indicator: VH backscatter value (dB)

Classification:

a single threshold (-18 dB) separates water and land

Output: binary (water/land)

Pros:

works through clouds, at night, and in poor weather

Cons: speckle noise; sensitive to threshold selection

Shoreline accuracy:

depends on threshold and image noise

Sentinel-2 (Optical, L2A)

Input: B03 (green), B04 (red), B08 (NIR), SCL

Indicators: NDWI (water), NDVI (vegetation)

Classification:

combination of multiple conditions (SCL + NDWI + NDVI)

Output: binary (water/land)

Pros:

better spectral separation (e.g., shallow water vs land)

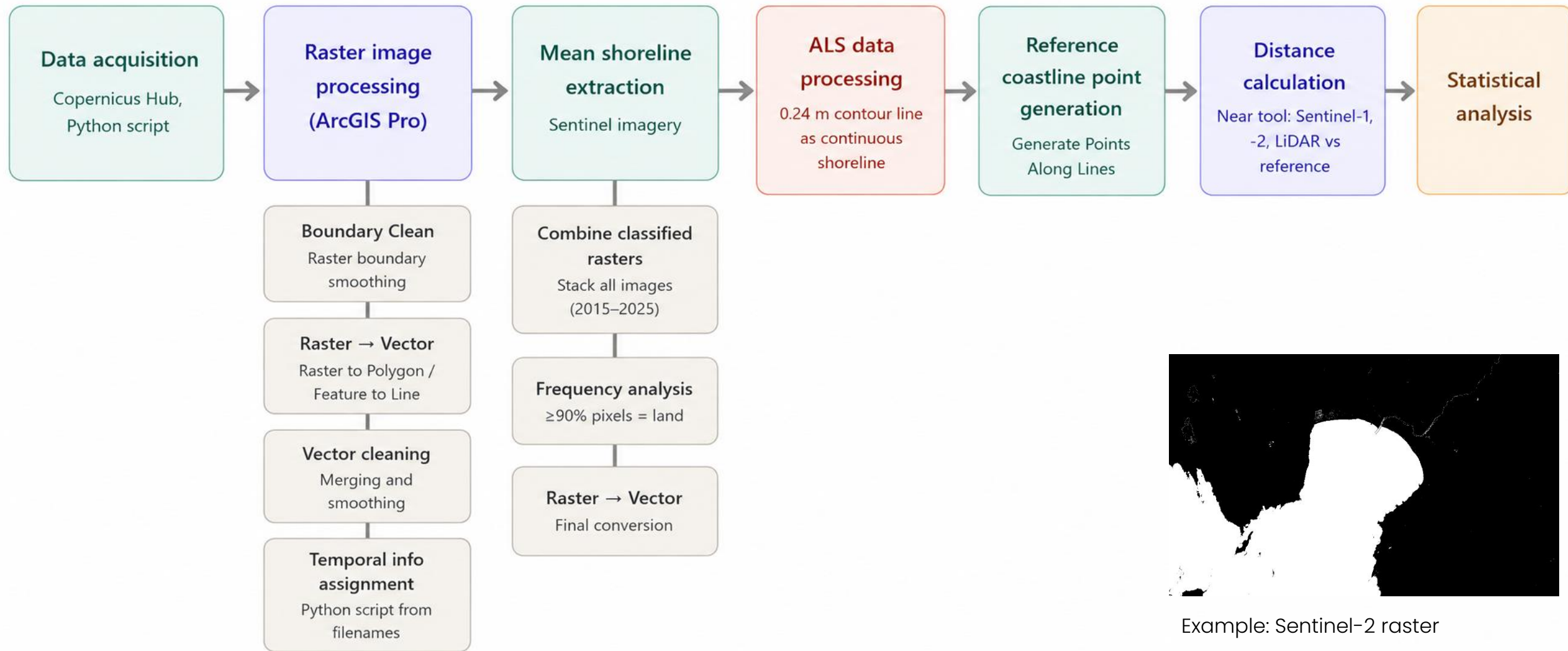
Cons: clouds, atmosphere, mixed shoreline pixels

Shoreline accuracy:

depends on index thresholds and SCL quality



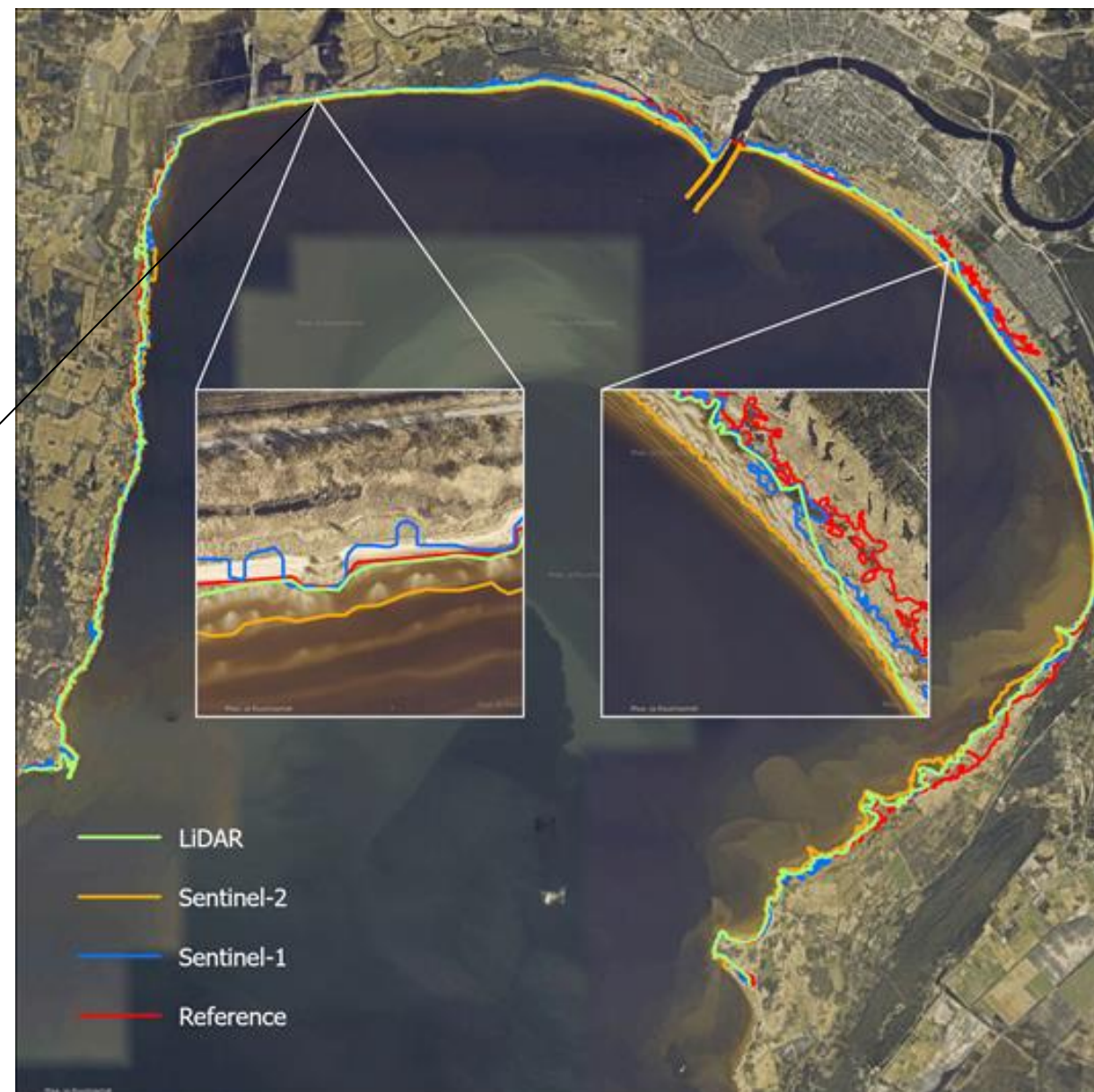
Methodology





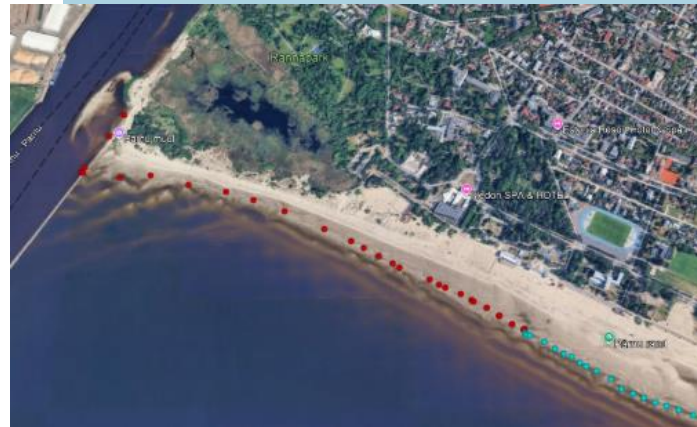
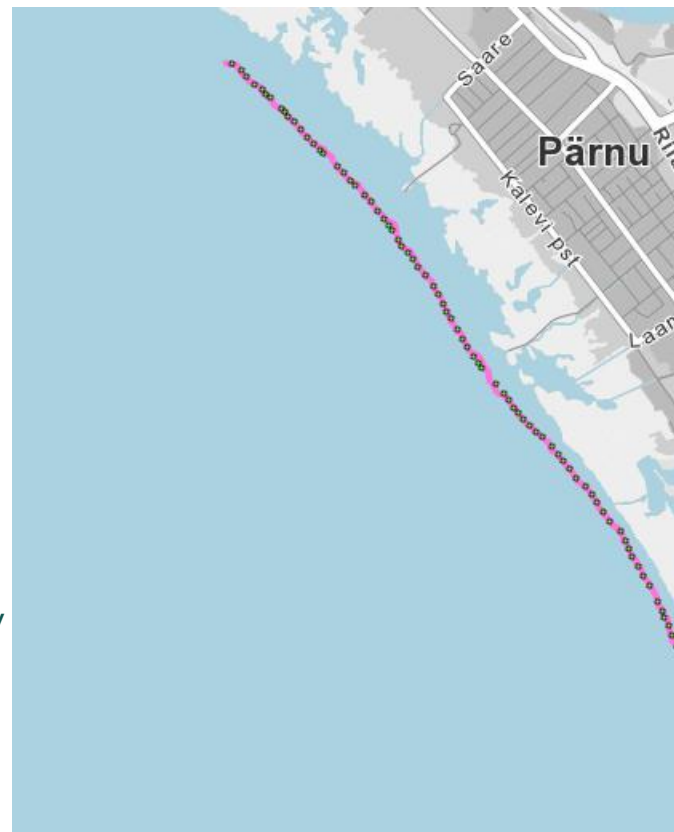
Results

Method	Min	Max	Mean	Std
LiDAR	0.0	351	36	46
Sentinel-1	0.0	359	40	44
Sentinel-2	0.3	372	82	66



Methodology Validation

- RTK GNSS field measurements (20 km), April 23, 2026
- Both Sentinel-1 and Sentinel-2 acquired imagery over the study area on the same day
- **Sentinel-2:** 6 ± 5 m
 - The mean deviation was below the spatial resolution (10 m)
- **Sentinel-1:** 130 ± 73 m



Field Name	Chart Preview	Min	Max	Mean	Std. Dev.	Median	Count	Range	IQR	Q1	Q3	CV	Skewness	Kurtosis
NEAR_DIST		0.024226	23.617216	5.976265	5.160802	4.565672	167 (100%)	23.59299	6.535112	2.212246	8.747358	0.86355	1.336877	4.657499



Future Works

- Testing different spectral indices
- Validate the methodology under different water-level conditions
- Validate the methodology across different coastline types
- Explore how satellite remote sensing data could be integrated into the national shoreline determination process





Second Case Study

World Championships Biathlon in Estonia 2027



Study area

- Area: 80 x 40 m
- *ca* 2400 m²
- Volume: 5 000 – 12 000 m³
- Twice a year from 2022



Field works

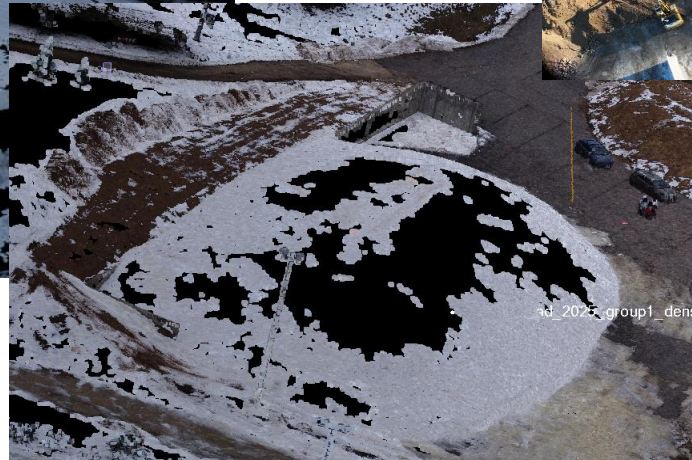
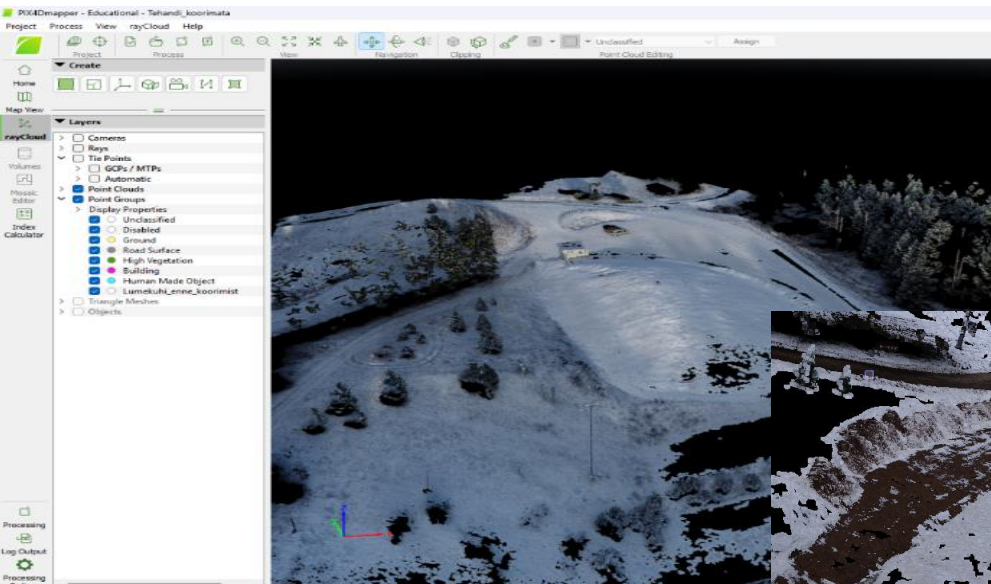
A base map was created for ground elevation modelling

- Drones: DJI Matrice 300, DJI Phantom 4 RTK, DJI Matrice 4E RTK
 - Altitude: 50 m
 - Camera tilt angle: 70°
 - Image overlap: 70% × 70%
 - Oblique imagery
- RTK GNSS (Trimble R8)



Data Processing

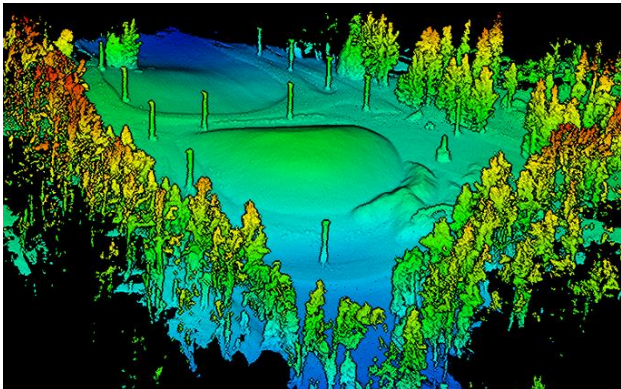
- Pix4D, AutoCAD Civil and Trimble Business Center





Volume Loss

Measurements	Volume (m ³)	Volume Loss (%)
March 7, 2023	10 931	26
December 4, 2023	8 031	
March 13, 2024	5 972	27
December 15, 2024	4 337	
March 5, 2025	8 950	31
January 16, 2026	6 159	
March 11, 2026	12 670	





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Thank You!

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Sentinel-1 ja 2 andmed

Sentinel-1 (SAR, VH kanal)	Sentinel-2 (optiline, L2A)
Andmetüüp: radar (SAR), ei sõltu pilvedest ega valgustingimustest	Andmetüüp: multispektraalne optiline (nähtav + NIR)
Sisend: VH polariseeritud tagasihajumine	Sisend: B03 (roheline), B04 (punane), B08 (NIR), SCL
Põhimõte: vesi annab madala tagasihajumise (tume radaripildil)	Põhimõte: vesi peegeldab rohelises rohkem kui NIR-is
Indikaator: VH dB väärtus	Indikaatorid: spektraalindeksid NDWI (vesi), NDVI (taimestik)
Klassifitseerimine: üks lävend (-18 dB) eraldab vee ja maa	Klassifitseerimine: mitme tingimuse kombinatsioon (SCL + NDWI + NDVI)
Tulemus: binaarne (vesi/maa)	Tulemus: binaarne (vesi/maa)
Tugevus: töötab pilvede, öö ja halva ilma korral	Tugevus: parem spektraalne eristus (nt madal vesi vs maa)
Puudused: müra (speckle), tundlik lävendi valikule	Puudused: pilved, atmosfäär, segapikslid rannas
Rannajoone täpsus: sõltub lävest ja mürast	Rannajoone täpsus: sõltub indeksite lävest ja SCL kvaliteedist